



HIGH PERFORMANCE COMPUTING 101

HPC Computing @ IU (Module 2)

Module 2 — Lesson 1

Background and Getting accounts



Overview

- Background and Prerequisites
- IU Compute Systems (RED, Quartz, Big Red 200)
- Selecting a System
- Creating Accounts and an RT Project



Background and Prerequisites

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Some Quick Background

- UITS Research Technologies responsible for all HPC systems
- All HPC systems live in the Bloomington & Indianapolis Data Centers
- UITS/RT teams support hardware, applications and training
- UITS Knowledge Base contains reference info & contact support info:
<https://kb.iu.edu/>



Assumed Background

- In this and the following module, we assume that you
 - have an IU Account username
 - have worked through the Duo multi-factor setup for IU services
 - have a working knowledge of Bash and basic Linux commands
 - this doesn't need to be extensive
- If you are a native Windows user, you may wish to install Windows Subsystem for Linux
 - See <https://learn.microsoft.com/en-us/windows/wsl/about>



HPC Compute Overview

- UITS currently hosts two supercomputers and an interactive desktop service
 - Quartz—High Throughput Computing cluster
 - Big Red 200—High Performance Cray supercomputer
 - Research Desktop (RED)—Virtual Linux desktop available 24/7
- These systems connect with
 - IU Science DMZ—networking optimized to support data
 - HPC Storage Systems—Slate and Geode file system groups
 - More on this in next module



IU Compute Systems

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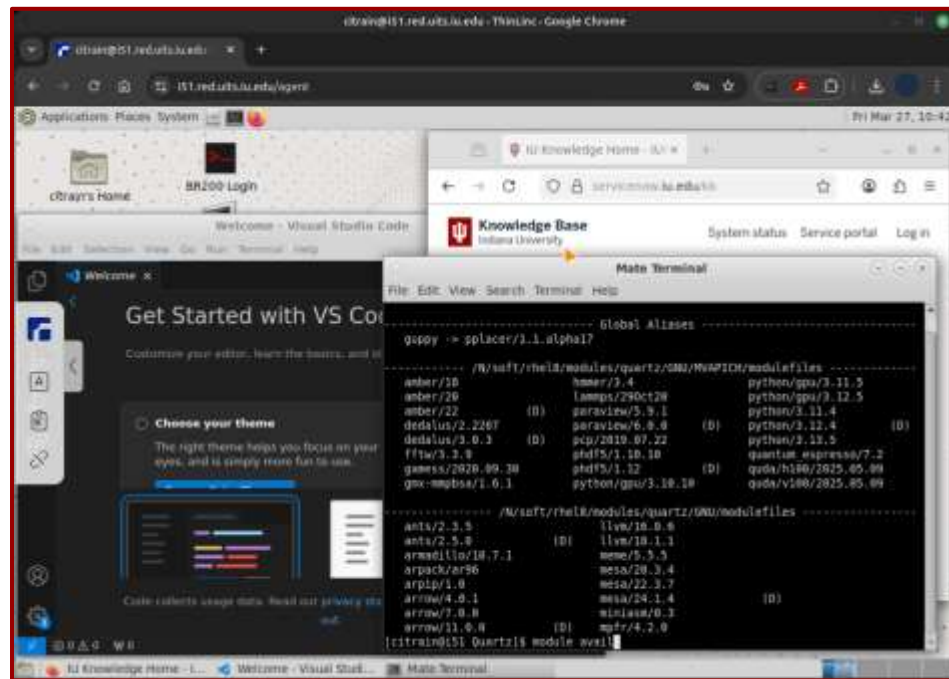


Research Desktop

- Interactive, graphical desktop on HPC platform
 - fully functioning Linux desktop with 100 GB RAM
 - many research applications available
 - local HPC filesystem access
- Provides an enduring windowing session
 - which one can disconnect from and return to
- Prepopulated with convenience functions and program icons
 - to access HPC machines; write and submit batch job requests

Research Desktop

- Web desktop extends w/
browser window
 - Note: small here, to fit on slide
- Each node supports multiple
users
 - users asked to keep apps to 4–8
cores max
- Desktop client available w/
greater functionality





Quartz

- Quartz is the workhorse system for general HPC work and AI
- Quartz contains CPU-only and GPU-accelerated nodes
- CPU-only partition contains 92 compute nodes
 - dual 64-core AMD EPYC 7742 2.25 GHz CPUs & 512 GB RAM
- GPU-accelerated nodes in two partitions
 - "gpu"—24 Apollo 6500 nodes
 - dual Intel Xeon 6248 2.5 GHz 20-core CPUs & 768 GB RAM
 - each node w/ 4 NVIDIA Tesla V100 PCIe 32 GB GPUs



Quartz

- GPU-accelerated nodes in two partitions
 - "hopper"—12 Dell XE8640 nodes
 - dual Intel Xeon 8468 2.1 GHz 48-core CPUs & 512 GB RAM
 - each node w/ 4 NVIDIA Hopper HGX H100 SXM 80 GB GPUs
- Each of the Quartz nodes has local SSD-backed storage in /tmp
 - hopper queue local disk is very high speed NVMe-based SSD
- All Quartz resources are scheduled by SLURM workload manager
 - more detail on this ahead



Big Red 200

- Big Red 200 is the IU flagship supercomputer
- An HPE Cray XE cluster with
 - 640 dual-cpu compute nodes
 - AMD EPYC 7742 2.25 GHz 64-core processors w/ 256 GB RAM
 - 64 gpu-accelerated, single-cpu nodes
 - AMD EPYC 7713 2.25 GHz 64-core processors w/ 512 GB RAM
 - each node w/ 4 NVIDIA Ampere A100 40 GB GPUs
- Includes the Cray Slingshot interconnect w/ Dragonfly topology
 - 200 Gbps per port with low latency



Big Red 200

- Big Red 200 is a state-of-the-art parallel supercomputer
 - designed to support both massive parallel computations and intensive AI workflows
- Like Quartz, the system has a very high throughput connection to the HPC file systems in the Data Center Research Pod
- Using Big Red 200 is similar to Quartz, via SLURM
 - requires a greater understanding of the system and development tools for success
 - starting with Quartz and moving to Big Red 200 is recommended



Selecting a System

- Background and Prerequisites
- IU Compute Systems (RED, Quartz, Big Red 200)
- **Selecting a System**
- Creating Accounts and an RT Project



Selecting a System

- Quartz and Big Red 200 have some similarities
 - but also some key differences
 - exist to support a range of differing HPC workflows
- Quartz is slightly more general
 - development and runtime tools may be more forgiving
 - just about any workflow on Quartz can run on Big Red 200 & vv
- If your workflow demands a high degree of parallelism, across server nodes, Big Red 200 should be your target



Selecting a System

- If you're working with AI, GPU specs will help you decide
- Need for large amounts of RAM may dictate your choices
- All nodes in both systems share access to same file systems
- Presently, all GPUs are NVIDIA
 - support CUDA, OpenACC and OpenCL programming
- UITS RT staff are happy to help you find the right place to start



Creating Accounts and an RT Project

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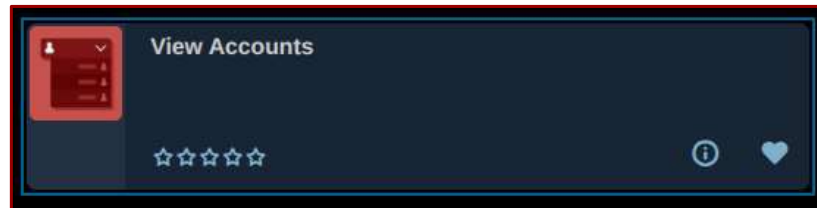
Creating Accounts

- We assume you have an IU Account username already
 - If not, go to <https://one.iu.edu/> and search for "Create My First IU Account"
- Accounts can be made via <https://one.iu.edu/> "View Accounts" app
 - or directly by browsing <https://access.iu.edu/Accounts>
- Quartz and Slate accounts provide usable storage
 - Home directory (100 GB)
 - Slate personal space (800 GB; expandable to 1600 GB)
- Start by creating Quartz and Slate accounts for yourself



Creating Accounts

- Direct your web browser to <https://one.iu.edu/>
- Sign in with your IU account
- Open the View Accounts application
- Select “Create Computing Account” on lower left of “Your Accounts” page
 - May be a delay of 1 hour or more after any account creation
 - Slate account requires an HPC account (Quartz or Big Red 200)



[Create Computing Account](#)



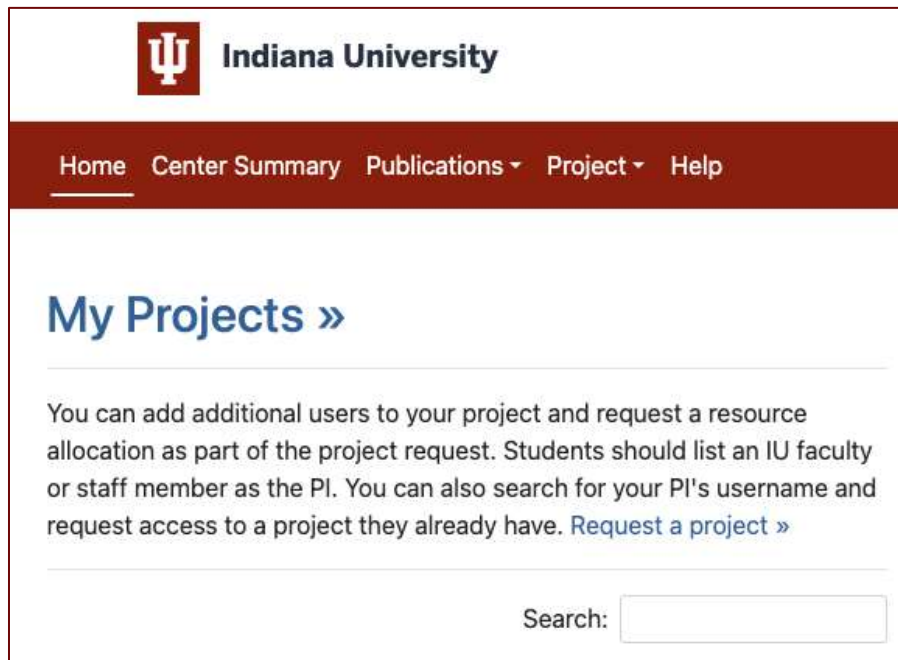
Creating an RT Project

- HPC compute and some storage systems are freely available to faculty, staff and students (with faculty/staff sponsorship)
 - Research Desktop should be available to students immediately after creating a Quartz account
- Research Technologies tracks the usage of some resources by using project accounts, needed to submit batch jobs
- After creating your compute accounts, you must create or join a project to utilize the batch queue



Creating an RT Project

- After creating your Quartz account, go to <https://projects.rt.iu.edu/>
- In the resulting web page, click "Log in" and authenticate to your account
- Then click "Request a project" as seen in image to the right, or go to
 - <https://projects.rt.iu.edu/project/create/>
- See "About RT Projects at IU" in KB for more info





Reflection

- You should now have an understanding of:
 - IU's HPC compute systems
 - Quartz and Slate account creation
 - RT Project creation
- Next lesson covers using your accounts:
 - Accessing RED and SSH into Quartz
 - Loading software with modules
 - Submitting your first SLURM job
- Be sure to complete the reflection for this lesson